

Show Your Work

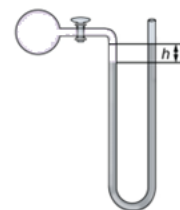
Quiz 7 May 1st 2025 Chem 1127

Name (First, Last): _____

Note the quiz is front & back + Equation Sheet/Periodic Table

Points: _____/16₁₈

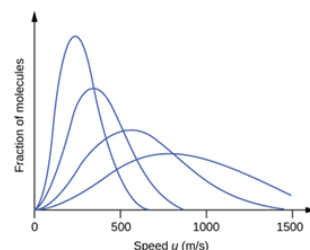
1. The pressure of a sample of gas is measured at sea level ($P_{\text{atm}} = 760 \text{ mm Hg}$) with an open-end Hg (mercury) manometer (shown here). If $h = 30 \text{ cm}$, determine the pressure of the gas in mm Hg. (1 pt)



- What has greater pressure, the **atmosphere** or the **gas**? Circle it (0.5 pt)
 - Identify where in the manometer the mercury (Hg) is. (0.5 pt)
2. Write the ideal gas law. (1.5 pt)
- This semester only a single experiment was done in lecture. It involved gas, explain how the gas's properties in the bottle changed, using the variables of the ideal gas law, and explain how it gave rise to the phenomenon that was observed between the egg & bottle. (1.5 pt)
3. If 2.50 g of SF_4 is introduced into an evacuated 500 mL container at 83°C , what pressure (in atm) is developed?(2 pt)
- What does evacuated or vacuum mean in the context of gases? Briefly define it. (0.5 pt)
4. What is the molar mass of a gas if 0.0494 g of the gas occupies a volume of 0.100 L at a temperature 26°C and a pressure of 0.404 atm? (2 pt)
5. Briefly define the term effusion. (1 pt)
6. Breaking chemical bonds is an **endothermic/exothermic** process, **releasing/absorbing/requiring** energy. Circle the correct bolded term in each pair. (1 pt)

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7. Identify the curve that corresponds to the gas of highest molar mass. briefly explain your answer. (1 pt)



- a. For two ideal gases with different molar mass at the same temperature, what is **different** between them? Circle it below. (0.5 pt)

Average Kinetic Energy

Average Velocity

8. We discussed a theoretical space-based weapon involving heavy tungsten cylinders, colloquially called 'Rods from god'. Discuss how this weapon would have kinetic and potential energy. (1.5 pt)

Kinetic Energy

Potential Energy

9. Consider if your hand is a system, you touch a red hot burner on your stovetop. Describe what the heat flow (q) is for your hand. (0.5 pt)

- a. Is the heat/energy solely used to raise your hand's temperature? Briefly discuss. (0.5 pt)

10. ~A12.27. In a coffee cup calorimeter, 0.800 g of ammonium nitrate(NH_4NO_3) is dissolved in 100.0 g of water. The water decreases in temperature from 21.11 to 20.39 °C. given that water has a specific heat capacity of $4.18 \text{ J g}^{-1} \text{ }^\circ\text{C}^{-1}$, what is the amount of heat, in joules, transferred in this reaction? (1.5 pt)

11. On the given image of the bomb calorimeter, identify

- where the sample is placed. (0.5 pt)
- Whether the sample chamber has constant volume or pressure. (0.5 pt)
- Where a temperature change for the **surroundings**, is relevant to your experiment. You may briefly discuss this (0.5 pt)



12. At the same T & P, what mass of H_2O_2 gas occupies the same volume as 20 mol He? (1.5 pt)

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Equations

$$1 \text{ cm} = 10 \text{ mm} \quad 1 \text{ kJ} = 1000 \text{ J}$$

$$1000 \text{ g} = 1 \text{ kg} \quad 1000 \text{ mL} = 1 \text{ L}$$

$$\text{Ideal Gas Law (NOT GIVEN)}$$

$$R = 0.0821 \text{ L} \cdot \text{atm} / (\text{mol} \cdot \text{K})$$

$$T_K = T_C + 273.15$$

$$KE = \frac{1}{2} m \cdot v^2$$

$$q = \text{mass} \cdot c \cdot \Delta T$$

$$\Delta T = T_{\text{fin}} - T_{\text{in}}$$

$$\frac{\text{effusion rate}_A}{\text{effusion rate}_B} = \frac{\sqrt{MM_B}}{\sqrt{MM_A}}$$

$$\frac{\text{time}_A}{\text{time}_B} = \left(\frac{MM_A}{MM_B} \right)^{1/2}$$

$$P_A + P_B + P_C + \dots = \Sigma_i P_i = P_{\text{Total}}$$

$$P_A = X_A \times P_{\text{Total}}$$

where

$$X_A = \frac{n_A}{n_{\text{Total}}}$$

Heat flow

$$q = \text{mass} \times c \times \Delta T$$

$$\Delta T = T_{\text{final}} - T_{\text{initial}}$$

Coffee-cup calorimeter

$$q_{\text{reaction}} = -\text{mass}_{\text{water}} \times 4.18 \text{ J/g} \cdot ^\circ\text{C} \times \Delta T$$

Bomb calorimeter

$$q_{\text{cal}} = C_{\text{cal}} \times \Delta T$$

Bomb calorimeter

$$q_{\text{reaction}} = -(q_{\text{cal}} + q_{\text{H}_2\text{O}})$$

Period Group

Periodic Table of the Elements

1 H hydrogen 1.008	2 He helium 4.003											13 B boron 10.81	14 C carbon 12.01	15 N nitrogen 14.01	16 O oxygen 16.00	17 F fluorine 19.00	18 Ne neon 20.18
3 Li lithium 6.94	4 Be beryllium 9.012											5 Al aluminum 26.98	6 Si silicon 28.09	7 P phosphorus 30.97	8 S sulfur 32.06	9 Cl chlorine 35.45	10 Ar argon 39.95
11 Na sodium 22.99	12 Mg magnesium 24.31											13 Al aluminum 26.98	14 Si silicon 28.09	15 P phosphorus 30.97	16 S sulfur 32.06	17 Cl chlorine 35.45	18 Ar argon 39.95
19 K potassium 39.10	20 Ca calcium 40.08	21 Sc scandium 44.96	22 Ti titanium 47.87	23 V vanadium 50.94	24 Cr chromium 52.00	25 Mn manganese 54.94	26 Fe iron 55.85	27 Co cobalt 58.93	28 Ni nickel 58.69	29 Cu copper 63.55	30 Zn zinc 65.38	31 Ga gallium 69.72	32 Ge germanium 72.63	33 As arsenic 74.92	34 Se selenium 78.97	35 Br bromine 79.90	36 Kr krypton 83.80
37 Rb rubidium 85.47	38 Sr strontium 87.62	39 Y yttrium 88.91	40 Zr zirconium 91.22	41 Nb niobium 92.91	42 Mo molybdenum 95.95	43 Tc technetium [97]	44 Ru ruthenium 101.1	45 Rh rhodium 102.9	46 Pd palladium 106.4	47 Ag silver 107.9	48 Cd cadmium 112.4	49 In indium 114.8	50 Sn tin 118.7	51 Sb antimony 121.8	52 Te tellurium 127.6	53 I iodine 126.9	54 Xe xenon 131.3
55 Cs cesium 132.9	56 Ba barium 137.3	57-71 La-Lu *	72 Hf hafnium 178.5	73 Ta tantalum 180.9	74 W tungsten 183.8	75 Re rhenium 186.2	76 Os osmium 190.2	77 Ir iridium 192.2	78 Pt platinum 195.1	79 Au gold 197.0	80 Hg mercury 200.6	81 Tl thallium 204.4	82 Pb lead 207.2	83 Bi bismuth 209.0	84 Po polonium [209]	85 At astatine [210]	86 Rn radon [222]
87 Fr francium [223]	88 Ra radium [226]	89-103 Ac-Lr **	104 Rf rutherfordium [267]	105 Db dubnium [270]	106 Sg seaborgium [271]	107 Bh bohrium [270]	108 Hs hassium [277]	109 Mt meitnerium [276]	110 Ds darmstadtium [281]	111 Rg roentgenium [282]	112 Cn copernicium [285]	113 Nh nihonium [285]	114 Fl flerovium [289]	115 Mc moscovium [288]	116 Lv livermorium [293]	117 Ts tennessine [294]	118 Og oganesson [294]

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57 La lanthanum 138.9	58 Ce cerium 140.1	59 Pr praseodymium 140.9	60 Nd neodymium 144.2	61 Pm promethium [145]	62 Sm samarium 150.4	63 Eu europium 152.0	64 Gd gadolinium 157.3	65 Tb terbium 158.9	66 Dy dysprosium 162.5	67 Ho holmium 164.9	68 Er erbium 167.3	69 Tm thulium 168.9	70 Yb ytterbium 173.1	71 Lu lutetium 175.0
89 Ac actinium [227]	90 Th thorium 232.0	91 Pa protactinium 231.0	92 U uranium 238.0	93 Np neptunium [237]	94 Pu plutonium [244]	95 Am americium [243]	96 Cm curium [247]	97 Bk berkelium [247]	98 Cf californium [251]	99 Es einsteinium [252]	100 Fm fermium [257]	101 Md mendelevium [258]	102 No nobelium [259]	103 Lr lawrencium [262]

Atomic number → 1

Symbol → **H**

Atomic mass → 1.008

Name → hydrogen

Metal	Solid
Metalloid	Liquid
Nonmetal	Gas